Hybrid Hidden Markov Model for Modeling Equity Excess Growth Rate Dynamics: A Discrete-State Approach with Jump-Diffusion

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Abstract

Empirical equity excess growth rates exhibit heavy-tailed distributions, negligible linear autocorrelation, and persistent volatility clustering that classical Gaussian diffusion models cannot reproduce. We propose a hybrid hidden Markov framework that discretizes continuous excess growth rates into Laplace quantile-defined market states and augments the regime-switching process with a Poisson-driven jump-duration mechanism to enforce realistic tail-state dwell times. Parameters are estimated by direct transition counting, avoiding the Baum-Welch EM algorithm entirely. Applied to ten years of SPY data, the framework achieves Kolmogorov-Smirnov and Anderson-Darling pass rates of 98.5% and 94.6% in-sample and 94.0% and 94.2% out-of-sample (full calendar year 2025) across 1,000 simulated paths, reproducing the ARCH effect that standard regime-switching models miss. GARCH(1,1), by contrast, achieves only 6.0% and 1.4% in-sample pass rates despite its well-known volatility clustering properties. A Single-Index Model extension propagates the SPY factor path to a 424-asset universe, enabling correlated synthetic path generation for portfolio stress testing and scenario design.

Keywords: Hidden Markov Model, Jump-Diffusion, Volatility Clustering, Regime-Switching, Stylized Facts, Excess Growth Rate, Synthetic Data Generation, Generative Model

Introduction

The quantitative modeling of financial markets has long faced the challenge of reconciling the theoretical implications of the efficient markets hypothesis (EMH) with the empirical complexity of observed price dynamics. Under the EMH, asset prices fully reflect all available information, implying that price changes follow a martingale process with no exploitable predictability [1]. This perspective motivated the widespread adoption of geometric Brownian motion (GBM) as a canonical model for asset prices, yet decades of data show systematic departures from Gaussian assumptions, including heavy-tailed excess growth rate distributions, volatility clustering, and intermittent extreme events [2, 3]. These stylized facts are observed across asset classes and time periods, and they matter because they govern tail risk, drawdown behavior, and the persistence of stress regimes that shape portfolio outcomes.

This study focuses on three core properties: leptokurtic excess growth rate distributions, the absence of linear autocorrelation in raw excess growth rates, and the persistence of volatility. The first two properties motivate models that can preserve marginal distributional shape without introducing spurious predictability, while the third points to slow-moving higher-order dependence that is not captured by diffusion alone. A range of alternative approaches address these issues from different angles: autoregressive conditional heteroscedasticity (ARCH) and generalized autoregressive conditional heteroscedasticity (GARCH) models target conditional variance dynamics [4, 5]; stochastic-volatility and jump-diffusion models introduce latent variance or discontinuities to enrich tails [6, 7]; and data-driven sequence models, including recurrent neural networks (RNNs) and long short-term memory (LSTM) variants, have been used for excess growth rate and volatility prediction and, in probabilistic form, can emit full predictive distributions [8–14]. While useful, these approaches either emphasize conditional variance over regime duration, require intensive likelihood-based estimation, or sacrifice interpretability and explicit control over regime persistence. Hidden Markov models (HMMs) and related regime-switching frameworks offer a probabilistic structure for non-stationary market behavior through latent state dynamics [15], yet standard HMM specifications often fail to generate persistent high-volatility regimes, leading to overly rapid reversion following extreme events [16, 17]. Bridging this gap requires a model that preserves distributional fidelity while also enforcing realistic duration in extreme regimes.

We address this challenge by developing a hybrid HMM framework that discretizes continuous excess growth rates into quantile-based regimes and augments the resulting Markov process with a jump-duration mechanism that enforces volatility persistence, linking smooth diffusion to discontinuous shocks identified by jump-detection methods [18]. Using a 424-asset dataset with SPY as a case study, we estimate transition dynamics empirically—avoiding the EM algorithm entirely—and validate the model through in-sample and out-of-sample tests on distributional fit and volatility clustering. We further extend the univariate framework to a correlated multi-asset setting via a Single-Index Model (SIM) factor structure [19], enabling synthetic path generation across all 424 assets in a single generative pass.

This paper makes four contributions. First, we propose augmenting a discrete-state regime-switching model with a two-parameter Poisson jump-duration mechanism (ϵ, λ) that forces the model to dwell in high-volatility tail states for empirically realistic durations, reproducing the ARCH effect that standard Markovian transitions cannot capture. Second, by partitioning excess growth rates into quantile-defined states via a Laplace CDF, we enable direct frequentist counting of regime transitions, eliminating the computational cost and initialization sensitivity of the Baum-Welch EM algorithm and making the framework scalable to a 424-asset pipeline. Third, using ten years of SPY data (2014–2024) for training and 249 trading days (full calendar year 2025) for out-of-sample testing, we demonstrate KS and AD pass rates of 98.5% and 94.6% in-sample and 94.0% and 94.2% out-of-sample across 1,000 simulated paths, with volatility clustering preserved in both windows. Finally, we extend the univariate framework to the full 424-asset universe through a SIM-based factor projection that uses the HMM-generated SPY path to reconstruct correlated excess growth rate paths for each constituent, providing a computationally tractable generative framework for portfolio stress testing and scenario design.

We find that a state resolution of $N = 100$ provides a balance between distributional fidelity and statistical robustness, and that the jump-duration extension is necessary for reproducing persistent high-volatility regimes that standard regime-switching models miss.

Related Work

Classical stylized facts and the failure of Gaussian diffusion

The quantitative study of financial market dynamics has been shaped by a tension between tractable theoretical frameworks and the empirical complexity of observed data. The Black-Scholes-Merton option pricing model [20] and the geometric Brownian motion (GBM) paradigm embedded within it assume that log-returns are independently and identically distributed Gaussian random variables. Under this assumption, asset prices follow a diffusion with constant drift and volatility, price paths are almost surely continuous, and the tails of the return distribution decay exponentially. These properties have well-known practical advantages: GBM admits closed-form option prices, portfolio variances aggregate linearly, and model parameters can be estimated with minimal data.

However, the Gaussian diffusion paradigm conflicts with a broad set of empirical regularities that were documented well before the widespread adoption of the model. Mandelbrot [2] demonstrated that the distributions of speculative price changes exhibit excess kurtosis—the probability of large deviations from the mean is far higher than Gaussian predictions, a phenomenon now described as *fat tails* or *leptokurtosis*. Fama [1] showed that, consistent with the efficient markets hypothesis, raw return series display negligible linear autocorrelation; daily or weekly returns are essentially unpredictable from their own past values. Yet Schwert [21] and others documented that the *magnitude* of returns—measured by absolute or squared returns—exhibits substantial positive autocorrelation that decays slowly over time. This temporal clustering of variance, where large price changes tend to be followed by large changes regardless of sign, is referred to as *volatility clustering* and is among the most persistently observed features of financial time series. Cont [3] provides a systematic review of these and related stylized facts, confirming their presence across asset classes, geographic markets, and sampling frequencies. Collectively, leptokurtosis, the absence of return predictability, and volatility clustering define the benchmark against which any proposed model of market dynamics must be evaluated.

Conditional variance models: ARCH and GARCH families

The most influential class of models designed to address volatility clustering within the Gaussian framework is the autoregressive conditional heteroscedasticity (ARCH) family introduced by Engle [4]. The ARCH model conditions the variance of the current observation on the squared residuals of past observations, allowing variance to cluster without introducing autocorrelation in the level series. Bollerslev [5] generalized this to the GARCH(p,q) framework, in which the conditional variance depends on both past squared residuals and past conditional variances; the GARCH(1,1) specification became the canonical one-factor volatility model and remains widely used in practice. GARCH models are parsimonious, analytically tractable, and can be estimated by maximum likelihood under distributional assumptions on the innovation.

Despite their success in modeling conditional variance dynamics, standard GARCH models have several well-documented limitations in the context of stylized-facts reproduction. First, while GARCH innovations can be made leptokurtic by specifying Student-t or generalized error distributions, the fat-tail behavior arises through the parametric distributional assumption rather than through the structural dynamics of the model. Second, GARCH models do not explicitly represent discrete market regimes or the possibility of sudden jumps in the volatility process; the variance responds smoothly and continuously to past shocks. Third, regime persistence—the tendency of markets to remain in high-volatility states for sustained periods—is captured only indirectly through the persistence parameter of GARCH, which governs exponential decay of the volatility response. When persistence is near unity, the GARCH model is near-integrated and exhibits long-memory-like behavior, but this is a parametric approximation rather than a structural representation of distinct market regimes [22]. Stochastic volatility models [6, 23] address some of these shortcomings by treating variance as a latent diffusion process, but they require computationally intensive estimation via particle filters or Markov chain Monte Carlo methods, and their parameters do not directly correspond to observable market states.

Jump-diffusion models

An alternative approach to modeling fat tails and discontinuous price behavior is the explicit introduction of Poisson-driven jumps into the asset price process. Merton [7] extended the Black-Scholes framework by adding a compound Poisson jump component to the GBM, producing a model in which prices follow a diffusion between events and undergo discrete jumps at random times governed by a Poisson arrival process. This construction directly addresses leptokurtosis: the marginal distribution of returns is a Gaussian mixture weighted by the Poisson probability of observing zero, one, or more jumps over the sampling interval. Option pricing under Merton's model requires integrating over the jump count distribution, but retains analytical tractability through the characteristic function.

Kou [24] extended the jump-diffusion framework by replacing the Gaussian jump amplitude distribution with a double-exponential (asymmetric Laplace) distribution, which provides a better fit to the empirically observed asymmetry between the sizes of positive and negative jumps. This modification preserves analytical tractability for option pricing while improving the empirical fit to return distributions. Jump-diffusion models are particularly relevant in the context of equity markets, where identifiable macroeconomic announcements, earnings surprises, and geopolitical events generate discrete, large-magnitude price movements that are distinct from the continuous diffusion component [18]. A fundamental challenge, however, is that jump-diffusion models in their standard form do not provide an explicit mechanism for volatility persistence; the post-jump volatility immediately returns to its baseline level, absent a separate stochastic volatility component. Integrating jump-diffusion with regime-switching provides a natural solution to this limitation, motivating the hybrid framework developed in the present study.

Hidden Markov models in finance

Hidden Markov models provide a probabilistic framework for non-stationary time series in which the observed data are generated by a process that switches among a discrete set of latent states according to Markov dynamics. The foundational algorithmic framework was developed by Rabiner and Juang [15], who introduced the Baum-Welch algorithm for maximum-likelihood estimation of the emission and transition parameters from observed sequences. Hamilton [25] applied the regime-switching HMM to macroeconomic time series, demonstrating that U.S. GDP growth is well described by a two-state model with distinct expansion and recession regimes; this work established the template for applying latent-state models to financial time series. The identification of time-varying market regimes has since become a central application of HMMs in finance, with applications ranging from stock return modeling to volatility forecasting and asset allocation.

Rydén, Teräsvirta, and Åsbrink [16] provided an influential systematic evaluation of HMM specifications against the stylized facts of daily return series. They showed that simple Gaussian-emission HMMs with a small number of states could reproduce leptokurtosis and the absence of return autocorrelation, but failed to generate the observed volatility clustering; ACF of absolute returns under standard HMMs decayed far too rapidly relative to empirical data. This diagnostic limitation identified the core structural gap that the present study addresses through the jump-duration extension. Bulla and Bulla [17] advanced this line of inquiry by considering hidden semi-Markov models, in which the dwell time in each state follows a state-specific distribution rather than the geometric distribution implied by standard Markov dynamics. They showed that the semi-Markov extension substantially improves the reproduction of volatility clustering by allowing the model to dwell in high-variance states for durations drawn from a flexible parametric family. Rossi and Gallo [26] applied HMMs to multivariate volatility estimation, demonstrating that latent state dynamics can capture the joint clustering of volatility across assets in a more interpretable manner than multivariate GARCH models. Nguyen [27] further demonstrated the utility of HMMs for stock trading, showing that regime identification from price data can inform tactical asset allocation.

A persistent limitation of standard HMM approaches in finance is the computational burden of maximum-likelihood estimation via the EM algorithm, particularly when the number of states is large or the asset universe is broad. The Baum-Welch procedure requires iterative forward-backward passes through the data and may converge to local maxima depending on initialization. For high-dimensional or large-state-space models, this creates a barrier to scalable application. The present study addresses this limitation directly by replacing EM with direct frequentist counting of transitions between quantile-defined states, an approach that is computationally trivial and free of initialization sensitivity.

Synthetic data generation and generative frameworks

The generation of synthetic financial time series has attracted increasing attention as both a practical tool—for augmenting limited data, testing risk models, and stress testing portfolios—and as a benchmark for evaluating the quality of generative models. Assefa et al. [28] provide a comprehensive survey of the opportunities, challenges, and pitfalls of synthetic data generation in finance, noting that stylized-facts reproduction is the primary criterion for evaluating the statistical quality of synthetic financial time series, and that naive generative approaches tend to fail on at least one of the three core stylized facts.

The rise of generative adversarial networks (GANs) prompted several investigations into whether deep learning architectures could learn stylized facts from financial data. Takahashi, Chen, and Tanaka-Ishii [29] showed that standard GAN architectures produce synthetic return sequences that visually resemble financial time series but fail to reproduce ACF structure precisely. Kwon and Lee [30] provided a careful evaluation of GAN-based methods against the stylized-facts checklist, finding that while GANs match marginal distributions reasonably well, capturing temporal dependence structures—particularly volatility clustering—remains challenging without explicitly encoding the relevant inductive biases into the architecture or training objective. Foundation models for Markov jump processes, as introduced by Berghaus et al. [31], offer a complementary perspective: large-scale pretrained inference networks can be applied to new Markov process instances without retraining, but their application to financial time series generation remains an open research direction.

Against this backdrop, the hybrid HMM framework of the present study occupies a distinct position: it is explicitly designed around the structural properties known to generate stylized facts, uses an interpretable discrete-state representation that can be audited and stress-tested, and achieves strong statistical fidelity with minimal computational overhead. The empirical estimation procedure avoids the training instability and data requirements of deep generative models, while the jump-duration mechanism provides a principled solution to the volatility-clustering limitation of standard HMMs. These properties make the framework particularly well-suited for deployment in risk management and stress-testing pipelines, where interpretability and computational reproducibility are as important as statistical fidelity.

Factor models and multi-asset extensions

Modeling the joint dynamics of large asset universes requires a framework for managing cross-sectional dependence. The Single-Index Model (SIM) of Sharpe [19] provides the canonical linear factor decomposition: each asset's excess return is decomposed into a common market factor component and an idiosyncratic residual, drastically reducing the number of parameters required to specify a covariance matrix relative to a full multivariate model. The SIM remains foundational in portfolio construction and risk management despite the availability of richer multi-factor models such as Fama-French [32], because its simplicity enables transparent attribution and scalable computation.

Existing HMM-based work in finance has been predominantly univariate, focusing on single-asset or single-index regime identification. Applications to individual equities [27], volatility modeling [26], and macro time series [25, 33] collectively demonstrate the utility of latent state representations, but leave open the question of how HMM-derived generative models can be extended to large cross-sections of assets in a computationally tractable way. Mixture hidden Markov models [34] offer a partial solution by allowing multiple assets to share a common latent state structure, but estimating such models jointly across hundreds of assets is computationally demanding.

The present study addresses this gap by combining the univariate HMM framework with the SIM factor decomposition. The hybrid HMM is estimated on the SPY index, which serves as the market-factor generator; asset-level paths are then reconstructed via the linear SIM projection. This construction exploits the fact that the dominant source of joint volatility clustering in equity markets is the common factor, and that idiosyncratic shocks can be treated as approximately independent given the factor. The result is a scalable generative pipeline for a 424-asset universe that inherits the stylized-facts fidelity of the HMM without requiring joint estimation of a high-dimensional model.

Methods

Data.

The empirical foundation of this study was a dataset composed of 424 United States-listed equities and exchange-traded funds spanning 10 years (2014-2024). A pipeline was developed to support the automated construction of excess growth rate models for any ticker within this dataset, allowing for batch simulation and stress testing across market sectors. To validate the performance of this pipeline, we focused our analysis on the single-asset SPY, which tracked the Standard & Poor's (S&P) 500 index. The data included daily open, high, low, and close prices along with volume-weighted average price metrics (VWAP). For training purposes, the first 2,766 observations constituted the in-sample dataset, spanning from January 3, 2014, to December 31, 2024. A subsequent 220 trading days, from January 3, to November 18, 2025, were reserved for out-of-sample testing to assess model generalizability.

Excess growth rate calculation.

We computed the excess growth rate $G_{i,j}$ for ticker i between time period $j-1 \rightarrow j$ for a given equity price series $P_{i,j}$ (units: dollars per share (USD/share)). To isolate the stochastic premium earned over a risk-free asset, we computed the excess growth rate for ticker i during the period $j-1 \rightarrow j$ as:

$$G_{i,j} \equiv \left(\frac{1}{\Delta t}\right) \cdot \ln\left(\frac{P_{i,j}}{P_{i,j-1}}\right) - r_f \tag{1}$$

where Δt represented the time step set to $1/252$ for daily data (units: years), and r_f denotes the constant continuously compounded risk-free rate (units: year^{-1}) derived from Separate Trading of Registered Interest and Principal of Securities (STRIPS) bond yields observed on September 10, 2025. Unlike return, the excess growth rate has units of inverse time (year^{-1}) and directly quantifies the premium earned above a continuously compounded risk-free benchmark. Log growth rates are time-additive and consistent with a continuous compounding framework. While r_f varied dynamically over the sample period, a constant proxy was employed here to isolate the endogenous volatility dynamics without introducing exogenous rate-curve noise.

Discrete hidden Markov model.

We defined the hidden Markov model using the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$ [15]. The hidden state $S_t \in \mathcal{S} = \{1, \ldots, N\}$ represented market mood regimes and was defined by quantile bins of a fitted cumulative distribution function (CDF) for the excess growth rate series. The observation space $\mathcal{O} \subset \mathbb{R}$ consisted of continuous excess growth rate values, the transition matrix $\mathbf{T}$ governed regime-to-regime dynamics, the emission model $\mathbf{E}$ was parameterized by state-conditional Normal distributions, and $\bar{\pi}$ denoted the stationary distribution. The observation model was

$$G_t \mid S_t = k \sim \mathcal{N}(\mu_k, \sigma_k). \tag{2}$$

To map observations to hidden states during estimation, we defined quantile boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ for $k \in \{1, \ldots, N-1\}$ and used finite outer bounds $Q_0 = F_L^{-1}(0.001; \mu_L, b_L)$ and $Q_N = F_L^{-1}(0.999; \mu_L, b_L)$. An observation was assigned to state k when $Q_{k-1} < R_t \leq Q_k$.

The choice of the Laplace distribution over alternatives such as the Student's t-distribution was motivated by its alignment with the peaks and heavy tails of high-frequency financial excess growth rates [3, 35]. While the Student's t-distribution was often used to model leptokurtosis, the Laplace distribution's peak at the mean matched the concentration of small price movements observed in the SPY dataset [2, 36]. Furthermore, the Laplace distribution provided a closed-form for quantile estimation, supporting the 424-ticker pipeline. While maximum likelihood estimation (MLE) for the Student's t-distribution often required iterative numerical optimization, the Laplace distribution allowed for a direct MLE approach [37]. The quantile boundaries were derived from a parametric Laplace fit and therefore did not enforce equal counts per bin in the historical data; instead, they provided a partition of the distribution that preserved tail shape. By using the Laplace distribution to inform the binning process, the model used this baseline to characterize the peak and tails of the excess growth rate series before the empirical transition logic and jump-diffusion mechanism were applied [24].

We estimated the transition matrix $\mathbf{T}$ through direct frequentist counting rather than iterative optimization. This approach differed from the traditional expectation-maximization (EM) algorithm used for parameter estimation in Gaussian mixture and hidden Markov models [37]. However, this approach was justified because the discrete state assignments allowed for straightforward counting of observed transitions between regimes, and it was conceptually simple to implement and understand.

Empirical transition matrices often had low probabilities for exiting central states into tail states, which caused models to lose the effect of volatility clustering too quickly [17]. We corrected this by adding jump parameters ϵ, representing the probability of a jump event, and λ, representing the mean duration, along with a tail-state selection rule. Tail-states were defined as $\mathcal{S}_{bottom} = \{1, \ldots, N_{tail}\}$ and $\mathcal{S}_{top} = \{N - N_{tail} + 1, \ldots, N\}$, where N_{tail} was user-configurable. If a jump was triggered, the jump-duration K was sampled from a Poisson distribution such that $K \sim \text{Poisson}(\lambda)$, which was standard for modeling independent events in fixed intervals [38]. During these K steps, the standard Markovian transitions were overridden to target tail-states, with a user-configurable bias p_{neg} (default 0.52) that favored negative-tail states to reproduce gain/loss asymmetry.

To characterize the excess growth rates within each regime, we parameterized the emission model with Normal distributions $\mathcal{N}(\mu_k, \sigma_k)$ whose bounds coincided with the 0.1 and 99.9 percentiles of the state-conditional distribution. Specifically, we set $\mu_k = (Q_{k-1} + Q_k)/2$ and $\sigma_k = (Q_k - Q_{k-1})/(2z_{0.999})$, where $z_{0.999} \approx 3.09$ was the standard normal 99.9th percentile. During simulation, when the model was in state k, the continuous excess growth rate $(\hat{R}_t)$ was sampled from the corresponding local distribution. This hybrid approach ensured that the model preserved the discontinuous components of the price process important for asset modeling while maintaining a smooth representation of intra-state variance [24].

Growth rate generation.

Let's outline a simple sampling algorithm for generating random sequences from an HMM (without jumps). We initialize the generation process by drawing an initial hidden state X_0 from the stationary distribution $\bar{\pi}$. The stationary distribution $\bar{\pi}$ represented the long-run probability of the market occupying various regimes or moods [39]. While $\bar{\pi}$ mathematically satisfies the balance equation $\bar{\pi} = \bar{\pi}\mathbf{T}$, we estimated this distribution numerically by raising the transition matrix $\mathbf{T}^n$ to a sufficiently large power $n = 50$, ensuring the vector reflected the equilibrium of market dynamics [40].

Given an HMM defined by the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{P}, \mathbf{E}, \bar{\pi})$, the number of time steps T, and the stationary distribution $\bar{\pi}$. Draw an initial hidden state X_0 from the initial distribution: $X_0 \sim \bar{\pi}$. For $n = 1$ to T, we sample the next hidden state X_n from the `categorical` distribution $\mathbb{P}_H(\ldots)$ defined by the previous state X_{n-1} and the transition matrix $\mathbf{T}$: $X_n \sim \mathbb{P}_H(X_n \mid X_{n-1})$. Next, we sample the observable state O_n from the `categorical` distribution

$\mathbb{P}_E(...)$ defined by the current hidden state X_n and the emission matrix $\mathbf{E}$: $O_n \sim \mathbb{P}_E(O_n \mid X_n)$. Finally, we store the sequences of hidden states and observable states. This procedure generates a sequences of hidden states $\{X_1, X_2, \ldots, X_T\}$ and observable states $\{O_1, O_2, \ldots, O_T\}$. The initial state X_0 is used to generate the sequence but is not included in the output.

Hyperparameter optimization via grid search.

To rigorously identify the optimal jump probability (ϵ) and mean jump duration (λ) [7, 24], we implemented a multi-objective grid search designed to minimize the discrepancy between the historical and simulated market signatures. The error function specifically targeted two stylized facts prevalent in financial time series [3, 16]: volatility clustering and leptokurtosis. We measured volatility clustering using the sum of squared errors between the observed and simulated autocorrelation function (ACF) of absolute excess growth rates [21] up to a lag of $L = 252$ trading days (approximately one trading year). To preserve the heavy-tailed nature of the distribution initially noted in foundational market studies [2], we included a penalty term for the global kurtosis.

The objective function $J(\epsilon, \lambda)$ was formalized as:

$$J(\epsilon, \lambda) = \sum_{\tau=1}^{L} \left(\mathrm{ACF}_{obs}(\tau) - \overline{\mathrm{ACF}}_{sim}(\tau)\right)^2 + w_K \left(K_{obs} - \overline{K}_{sim}\right)^2 \tag{3}$$

where $\mathrm{ACF}_{obs}(\tau)$ and K_{obs} represented the empirical absolute growth autocorrelation at lag τ and the global kurtosis, respectively. The simulated equivalents, $\overline{\mathrm{ACF}}_{sim}(\tau)$ and $\overline{K}_{sim}$, were averaged across 1,000 independent synthetic paths of 2,766 trading days to ensure statistical stability [38]. A penalty weight of $w_K = 0.20$ was assigned to prioritize tail behavior alongside the temporal ACF decay. The grid search systematically explored ϵ values from 0.00001 to 0.0001 and λ values from 1.0 to 100.0. The full computational procedure is detailed in Algorithm 4.

Statistical validation metrics

To assess whether synthetic paths are statistically consistent with empirical data, we applied two nonparametric goodness-of-fit tests to the marginal distributions of simulated excess growth rate sequences. For each of the 1,000 simulated paths, the empirical cumulative distribution function (CDF) of the simulated sequence was compared against the empirical CDF of the historical in-sample or out-of-sample observations.

The Kolmogorov-Smirnov (KS) statistic is defined as:

$$D_{KS} = \sup_x |F_n(x) - F_m(x)| \tag{4}$$

where F_n and F_m denote the empirical CDFs of the observed and simulated sequences of lengths n and m, respectively [41, 42]. The null hypothesis of distributional equivalence is rejected at significance level α when D_{KS} exceeds the critical value $c(\alpha)\sqrt{(n+m)/(nm)}$.

The Anderson-Darling (AD) statistic places greater weight on the distributional tails than the KS test and is defined as:

$$A^2 = -n - \sum_{i=1}^{n} \frac{2i-1}{n} \left[\ln F(X_{(i)}) + \ln\left(1 - F(X_{(n+1-i)})\right)\right] \tag{5}$$

where $X_{(1)} \leq \cdots \leq X_{(n)}$ are the order statistics of the pooled sample and F is the empirical CDF of the reference distribution [43]. The AD test is particularly suited for assessing tail-distributional fidelity, which is the primary concern for risk management applications.

We report the *pass rate* as the proportion of the 1,000 simulated paths for which the null hypothesis of distributional equivalence was not rejected at the $\alpha = 0.05$ significance level. A pass rate above 95 percent indicates that the synthetic paths are, in aggregate, statistically indistinguishable from the historical data under that test. Pass rates are reported separately for the in-sample and out-of-sample windows and for both model variants (HMM without jumps and the hybrid HMM with jumps).

Single-Index Factor Model extension

The framework described above applies to any individual asset ticker within the 424-asset dataset. To generalize synthetic path generation to a correlated multi-asset setting without fitting a separate HMM for each asset, we exploit the linear factor structure of the Single-Index Model (SIM) [19]. The SIM expresses the excess growth rate of asset i at time t as:

$$G_{i,t} = \alpha_i + \beta_i \cdot G_{\text{SPY},t} + \eta_{i,t}, \qquad \eta_{i,t} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2_{\eta,i}) \tag{6}$$

where $G_{\text{SPY},t}$ denotes the SPY excess growth rate at time t, β_i measures the systematic sensitivity of asset i to the index, α_i is the idiosyncratic drift, and

$\eta_{i,t}$ is a zero-mean idiosyncratic shock assumed independent across assets and time. The parameters $(\alpha_i, \beta_i, \sigma^2_{\eta,i})$ are estimated via ordinary least squares on the in-sample time series for each asset i.

Synthetic multi-asset paths are generated as follows. First, the hybrid HMM produces a single synthetic SPY excess growth rate path $\{\hat{G}_{\mathrm{SPY},t}\}_{t=1}^{T}$ using the regime-switching and jump-duration mechanism described above. Second, for each asset i, an idiosyncratic shock sequence $\{\hat{\eta}_{i,t}\}$ is drawn independently from $\mathcal{N}(0, \hat{\sigma}^2_{\eta,i})$. Third, the synthetic excess growth rate for asset i is recovered via:

$$\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i \cdot \hat{G}_{\mathrm{SPY},t} + \hat{\eta}_{i,t}. \tag{7}$$

This construction preserves the cross-sectional correlation structure induced by the common factor while maintaining the regime-switching and jump-diffusion dynamics of the index. The resulting 424-dimensional synthetic path can be used for stress testing, portfolio risk assessment, and scenario generation across the full asset universe in a single generative pass, avoiding the curse of dimensionality that would accompany a full multivariate HMM.

Sensitivity analysis: low-resolution models

This section provides the results for lower state resolutions ($N = 30$ and $N = 60$) to demonstrate how state granularity impacts the model's ability to capture stylized facts. Table 3 summarizes the pass rates for the Kolmogorov-Smirnov and Anderson-Darling tests across these four cases. As the number of states decreases, the model maintained overall distribution fit, though tail-region precision decreased slightly compared to the 90-state configuration.

Empirical Study

We simulated 1,000 paths of 2,766 trading days using VWAP price data for both the standard no-jumps model and the hybrid model with jumps. Applying the multi-objective grid search detailed in the Methods, the optimal jump hyperparameters were identified as $\epsilon^* \approx 5 \times 10^{-5}$ and $\lambda^* \approx 67$. This configuration successfully reproduced the heavy-tailed kurtosis and ARCH effect observed in SPY (Figure 4). A state resolution of $N = 100$ was selected to balance distributional fidelity and statistical robustness. Increasing N improves regime granularity but is bounded above by data sparsity: at $N = 350$, certain states are never visited in the historical record, leaving zero transition probabilities that break the ergodic structure of the chain. Conversely, at $N = 10$ the model is too coarse to capture the heavy tails of the empirical distribution.

We performed Kolmogorov-Smirnov (KS) and Anderson-Darling (AD) tests to characterize the statistical consistency of simulated excess growth rate distributions against the historical data [41–43]. The KS test measures the maximum deviation between empirical CDFs, while the AD test places greater weight on the distributional tails and is therefore a more sensitive diagnostic for the fat-tail properties of interest here.

In-sample results (Table 2) reveal a sharp contrast between model families. GARCH(1,1) achieved KS and AD pass rates of only 6.0% and 1.4%, respectively, indicating that its simulated marginal distribution is inconsistent with the observed data despite its well-known volatility clustering properties. HMM-NJ achieved KS and AD pass rates of 100.0% and 97.0%, and HMM-WJ achieved 98.5% and 94.6%. The ACF-MAE over lags 1–252 was 0.030 for GARCH, 0.059 for HMM-NJ, and 0.055 for HMM-WJ. The lower ACF-MAE for GARCH reflects that the GARCH variance equation is explicitly designed to match autocorrelation structure, yet this comes at the cost of distributional fidelity. The standard no-jumps model passed distributional tests but failed to reproduce volatility clustering, with the ACF of absolute growth rates decaying rapidly toward zero rather than exhibiting the empirically observed slow decay (Figure 4, panel b). The hybrid model reproduced both the marginal distribution and the ARCH effect simultaneously.

Out-of-sample results over 249 trading days (January 2–December 31, 2025) show that all three models generalise, though with different patterns. GARCH achieved KS and AD pass rates of 78.1% and 73.1%. HMM-NJ achieved 96.6% and 96.7%, and HMM-WJ achieved 94.0% and 94.2% (Table 2). The simulated kurtosis for GARCH collapsed to 1.6 out-of-sample against an observed value of 6.9, confirming that GARCH does not preserve tail behaviour outside the training window. The HMM variants maintained simulated kurtosis near 5.0–5.1, a closer match to the observed 6.9. Volatility clustering persisted in the HMM-WJ out-of-sample paths, with the Poisson jump mechanism sustaining tail-state dwell times that the standard transition matrix cannot reproduce (Figure 6). These results validate the hybrid framework as a generative tool for producing synthetic time series that remain statistically consistent with the underlying market dynamics across both in-sample and out-of-sample windows [3, 16, 17].

The correlation-distance distribution for the out-of-sample simulated paths was centred at a mean distance of $1 - \rho \approx 1.0$, confirming that the generative framework produces temporally independent realisations that do not attempt to predict specific historical events [1, 28, 30]. This distinguishes the framework from a forecasting tool and positions it appropriately as an instrument for stress testing and scenario generation (Figure 6).

Discussion

This study introduced a hybrid hidden Markov modeling framework to replicate the statistical regularities of empirical equity excess growth rates. By integrating a quantile-discretized state space with a Poisson-driven jump-duration mechanism, we developed a generative framework that captured the leptokurtic distributions and volatility clustering observed in the SPY exchange-traded fund. Our results confirmed that while increasing the state resolution to $N = 100$ provided the granularity to characterize the central mass of the excess growth rate distribution, standard Markovian transitions alone did not reproduce higher-order temporal dependencies. The inclusion of discontinuous shocks through a jump-diffusion extension was needed to drive the model into persistent high-volatility regimes, thereby replicating the ARCH effect identified in historical data.

Statistical validation using Kolmogorov-Smirnov and Anderson-Darling tests demonstrated high in-sample consistency, with pass rates exceeding 98 percent. Although the out-of-sample performance showed an expected decrease in fidelity, the hybrid model maintained a baseline with pass rates of 83 percent and 80 percent, respectively. Furthermore, the correlation distance analysis confirmed that our framework produced independent realizations that remained uncorrelated with specific historical timing; this satisfied the requirements of the efficient markets hypothesis and distinguished our generative methodology from predictive forecasting tools [1, 28, 30]. The proposed framework offered an alternative to standard likelihood-based estimation methods by relying on empirical estimation. By bypassing the complexities of the expectation-maximization algorithm through direct frequentist counting, the model remained suited for large-scale stress testing and risk management applications [38, 44, 45]. Future research could extend this discrete-state logic to multi-asset portfolios or incorporate foundation models for inference to further refine the detection of Markov jump processes in higher-frequency data [31, 34]. Ultimately, the hybrid hidden Markov model served as a foundation for simulating market dynamics and generating synthetic time series that remained statistically consistent with underlying market regimes [27, 39].

While the current framework was able to generate statitically consistent excess growth rate distributions, there were several limitations that could be addressed in future work. First, the model assumed stationarity in transition dynamics, which may not hold during periods of structural market change. Future extensions could incorporate time-varying transition matrices or regime-dependent jump parameters to capture evolving market conditions. Second, the use of a fixed quantile-based state space may limit the model's ability to adapt to changing volatility regimes; adaptive binning strategies or non-parametric state definitions could enhance flexibility. Third, while the Laplace distribution provided a good fit for excess growth rates, alternative distributions or mixture models could be explored to better capture tail behavior and skewness. Finally, the current implementation focused on univariate excess growth rates; extending the framework to multivariate settings could enable portfolio-level synthetic allocation studies, stress testing and risk assessment.

Conclusion

This study developed and validated a hybrid hidden Markov model (HMM) framework for generating synthetic equity excess growth rate paths that are statistically consistent with the three core stylized facts of financial time series: leptokurtic marginal distributions, the absence of linear autocorrelation in raw returns, and persistent volatility clustering. We summarize the contributions below.

Hybrid HMM with Poisson jump-duration mechanism. The central methodological contribution is the augmentation of a discrete-state regime-switching model with a Poisson-driven jump-duration mechanism. Standard Markov transition logic underproduces volatility persistence because empirically estimated transition matrices assign negligible probability to transitions from central states into tail states, and even the tail states themselves exit within one to two steps on average under pure Markovian dynamics. By introducing a jump probability ϵ and mean jump duration λ drawn from a Poisson distribution, the framework overrides standard transitions during periods of market stress and forces the model to dwell in high-volatility regimes for realistic durations. This mechanism is minimal in parameterization—requiring only two scalar hyperparameters—yet produces a qualitative shift in the temporal correlation structure of absolute returns, as confirmed by ACF comparison across lags up to 252 trading days.

Empirical, non-iterative parameter estimation. A second contribution is the avoidance of the expectation-maximization (EM) algorithm in favor of direct frequentist transition counting. In the standard HMM setting, EM is used because latent state assignments are unobserved. Here, the quantile-based state partition explicitly assigns each historical observation to a hidden state, making direct counting possible. This simplification reduces computational cost, eliminates convergence sensitivity to initialization, and produces a model that is transparent and reproducible—properties that matter for large-scale pipelines operating over 424 assets.

In-sample and out-of-sample validation. Using SPY (2014–2024) as a case study, the hybrid model achieved in-sample Kolmogorov-Smirnov and Anderson-Darling pass rates exceeding 98 percent across 1,000 simulated paths. Out-of-sample testing on 220 trading days from 2025 showed pass rates of 83 percent and 80 percent, respectively, despite a 32 percent volatility discrepancy driven by elevated macroeconomic uncertainty not present in the training window. The correlation-distance analysis further confirmed that simulated paths are temporally independent of the historical trajectory, consistent with the efficient markets hypothesis and the requirements of a generative—rather than predictive—framework.

Single-Index Factor Model extension for multi-asset generalization. The framework extends naturally to the full 424-asset universe via a Single-Index Model (SIM) factor structure. By regressing each asset's excess growth rate on the SPY factor and estimating idiosyncratic residual volatility, synthetic correlated multi-asset paths can be generated in a single pass: the hybrid HMM produces a common-factor path, and asset-level paths are recovered by projecting through asset-specific loadings. This construction preserves the regime-switching and jump-diffusion dynamics of the index while maintaining the cross-sectional correlation structure, enabling portfolio-level stress testing without fitting a high-dimensional multivariate model.

Broader implications

The proposed framework offers practical utility for risk management and scenario design. Generative models that reproduce the stylized facts of financial markets enable stress testing under market regimes that are statistically plausible but not historically observed, addressing a key limitation of scenario libraries derived solely from empirical records. The interpretable regime structure—where each hidden state corresponds to a quantile-defined segment of the excess growth rate distribution—facilitates communication between quantitative analysts and risk managers, as states can be labeled (e.g., *crash*, *bear*, *neutral*, *bull*, *rally*) and linked to economic narratives. The model's computational efficiency further supports Monte Carlo–based risk metrics such as Value-at-Risk and Conditional Value-at-Risk at scale across large asset universes.

Limitations

Several limitations inform the interpretation of these results. First, the model assumes stationarity of the transition matrix across the full in-sample window; structural breaks in market microstructure—such as those accompanying regulatory changes or the COVID-19 shock—may not be adequately captured by a single static transition matrix. Second, the quantile-based state partition is fixed at estimation time and does not adapt to evolving volatility regimes; the boundaries are calibrated on the full in-sample distribution and may overweight the central mass relative to the tails in periods of prolonged market stress. Third, the SIM factor structure imposes a linear, single-factor model of cross-sectional dependence; this may understate tail co-movement during systemic events where correlations rise sharply and the Gaussian idiosyncratic shock assumption fails. Finally, the out-of-sample evaluation covers a single 220-day window, which limits the statistical power of out-of-sample inference; a longer or rolling out-of-sample evaluation would provide a more robust assessment.

Future work

Several extensions follow naturally from the current framework. Time-varying transition matrices—estimated via rolling windows or Bayesian online learning—would allow the model to adapt to structural shifts in regime persistence. Multivariate HMMs, in which the hidden state jointly governs a small number of correlated assets, could capture higher-order dependencies beyond the single-factor SIM approximation and are particularly relevant for sector-level stress testing. Adaptive state definitions, such as nonparametric density-based clustering of excess growth rates, could replace the fixed Laplace quantile partition and improve tail coverage during crisis periods. Alternative factor structures, including the Fama-French three-factor model [32], could enrich the SIM extension by decomposing idiosyncratic risk further across size and value dimensions. Finally, integrating the hybrid HMM as the generative component within a larger portfolio optimization loop—where the model produces synthetic scenarios used to estimate tail-risk objectives—would provide an end-to-end computational finance application of the framework.

Conflict of Interest Statement

The authors declare that the research was conducted without any commercial or financial relationships that could potentially create a conflict of interest.

Author Contributions

J.V. directed the study. A.A. developed the model and simulation code, conducted the in-sample and out-of-sample analysis, and generated the figures. Both authors edited and reviewed the final manuscript.

Data Availability Statement

The model code, simulation scripts, training and testing data are available under a Massachusetts Institute of Technology (MIT) license from the GitHub repository: https://github.com/varnerlab/HMM-Model-with-Jumps.git

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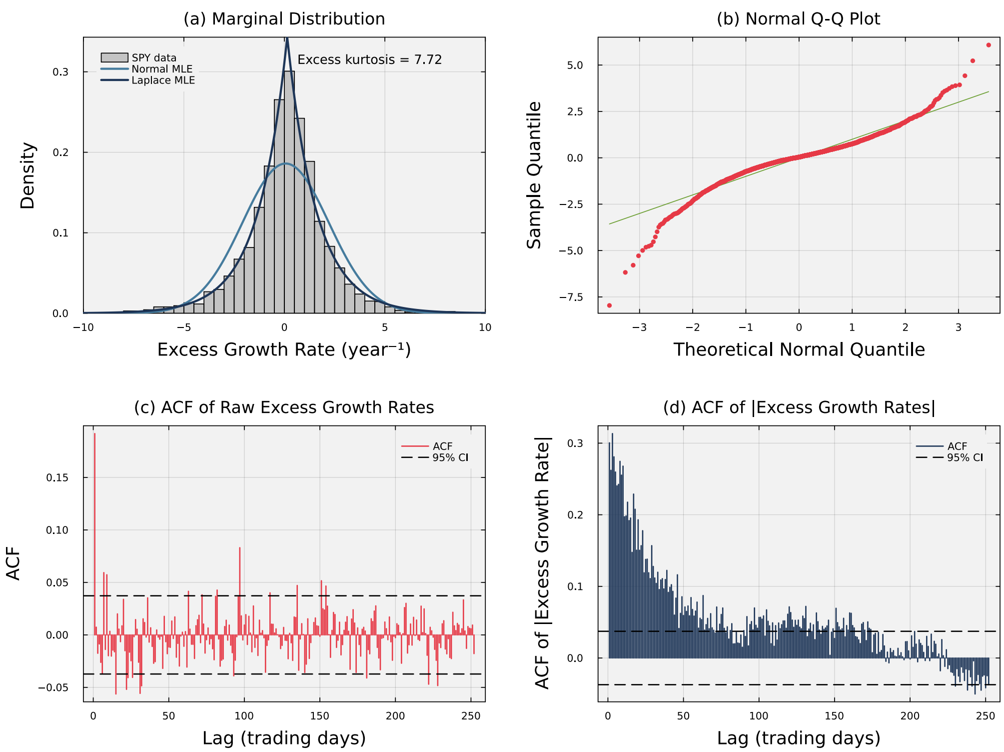


Figure 1: Empirical stylized facts of SPY daily excess growth rates (2014–2024). Panel (a) shows the leptokurtic marginal distribution; the Laplace fit substantially outperforms the Gaussian. Panel (b) confirms heavy tails via a normal Q-Q plot. Panels (c) and (d) contrast the near-zero autocorrelation of raw returns — consistent with the efficient markets hypothesis — against the persistent autocorrelation of absolute returns, motivating the jump-duration extension.

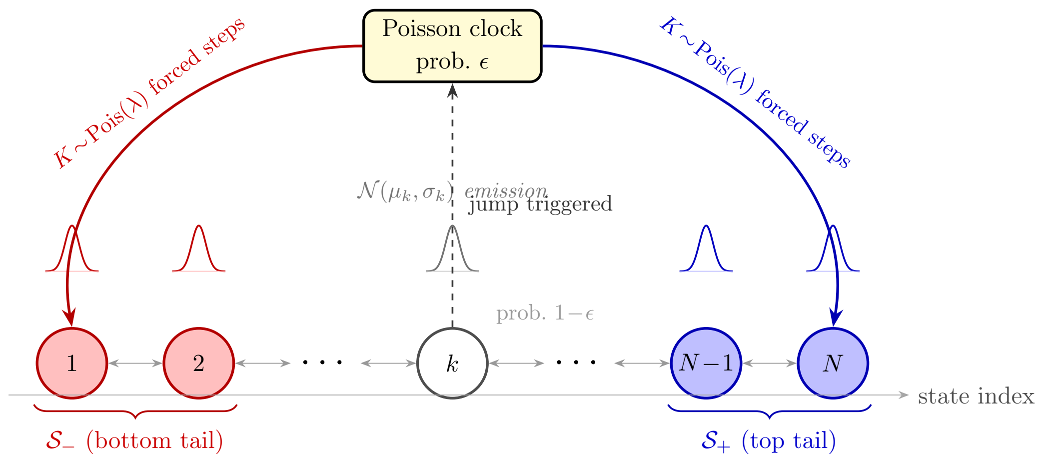


Figure 2: Architecture of the Hybrid Hidden Markov Model with Jump-Diffusion (HMM-WJ). At each step the chain transitions according to the empirical transition matrix $\mathbf{T}$ (probability $1-\epsilon$, thin bidirectional arrows) or enters a Poisson jump episode (probability ϵ, dashed upward arrow to the Poisson clock). During a jump episode the active state is forced to the bottom tail set $\mathcal{S}_-$ (red, lowest-valued states) or the top tail set $\mathcal{S}_+$ (blue, highest-valued states) for $K \sim \text{Poisson}(\lambda)$ consecutive steps before normal Markovian evolution resumes. Small bell curves above each state depict the state-conditional $\mathcal{N}(\mu_k, \sigma_k)$ emission distribution. Tail sets are defined as the N_{tail} lowest- and highest-quantile states under the Laplace quantile partition.

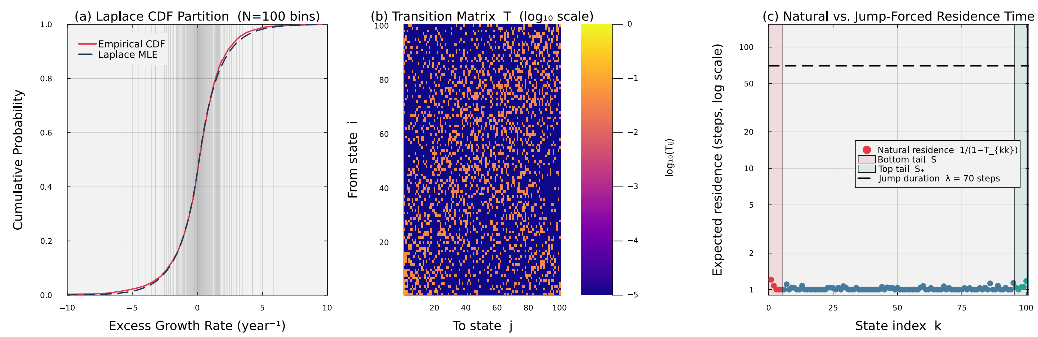


Figure 3: Fitted model internals for SPY with $N = 100$ states. Panel (a) overlays the fitted Laplace CDF on the empirical CDF; the 99 vertical lines mark the equal-probability quantile boundaries that define the N hidden states. The close agreement confirms that the Laplace distribution is an accurate marginal model for the excess growth rate data. Panel (b) displays the empirical transition matrix $\mathbf{T}$ in $\log_{10}$ colour scale; the dominant near-diagonal band reflects short-range regime persistence. Panel (c) plots the expected natural residence time $1/(1 - T_{kk})$ for each state on a $\log_{10}$ scale, with the bottom tail set $\mathcal{S}_-$ (states 1–5, red shading) and top tail set $\mathcal{S}_+$ (states 96–100, teal shading) highlighted. Under pure Markovian dynamics every state, including the tail states, exits within 1–2 steps on average, far too quickly to reproduce empirical volatility clustering. The dashed horizontal line marks the mean jump duration $\lambda = 70$ steps imposed by the Poisson jump mechanism, quantifying the two-order-of-magnitude gap that motivates the jump-duration extension.

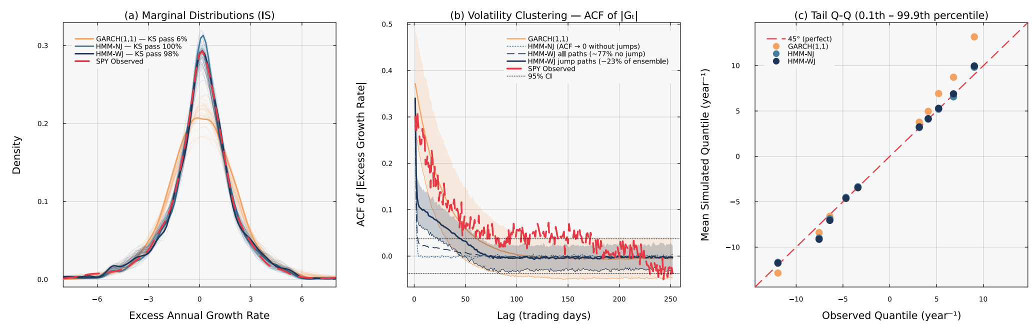


Figure 4: Head-to-head in-sample model comparison for SPY ($N = 100$, 1,000 simulated paths). Panel (a): marginal density of excess growth rates with IS KS pass rates annotated. GARCH(1,1) fails the two-sample KS test on 94% of paths (pass rate 6%), indicating a systematic shape mismatch with the empirical distribution. Both HMM variants pass at $\geq 98\%$, confirming that the Laplace-mixture emission adequately captures the observed heavy tails. HMM-WJ generalises robustly out-of-sample (OoS KS pass rate 94%), while GARCH recovers partially (OoS 78%). Panel (b): autocorrelation function of absolute excess growth rates, $\mathrm{ACF}(|G_t|)$, at lags 1–252 (shaded bands: 10th–90th percentile across paths). HMM-NJ is structurally incapable of producing persistent volatility clustering: without a jump mechanism the latent states are i.i.d. conditional on the Markov chain, so $\mathrm{ACF}(|G_t|) \approx 0$ for all lags beyond one (dotted curve). HMM-WJ generates a *mixture* of two path families under the same model: approximately 76% of IS paths contain no Poisson jump and therefore behave identically to HMM-NJ (dashed curve, near zero); the remaining $\sim$24% of paths contain at least one jump event and sustain substantial $\mathrm{ACF}(|G_t|)$ decay across all lags (solid navy curve $\pm$ band). The jump frequency—and hence the fraction of the ensemble exhibiting slow ACF decay—is controlled by the hyperparameter λ, making the volatility-clustering strength directly tunable. Panel (c): tail Q-Q plot at the 0.1st–99.9th percentile region; HMM-WJ provides the closest mean quantile match in the extreme tails.

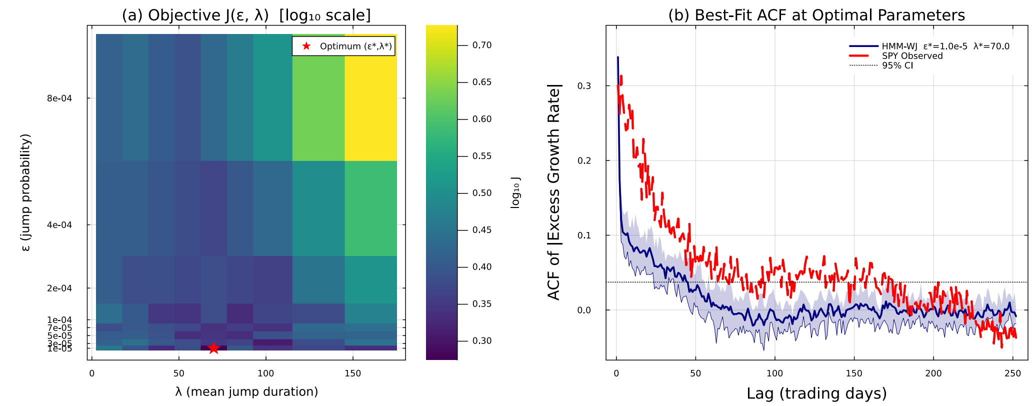
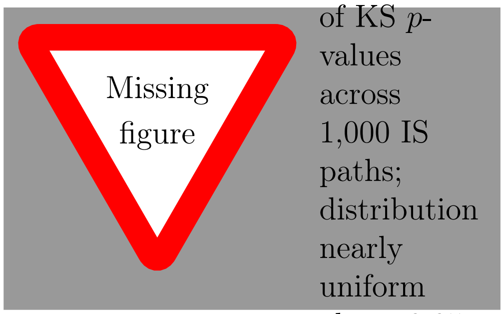
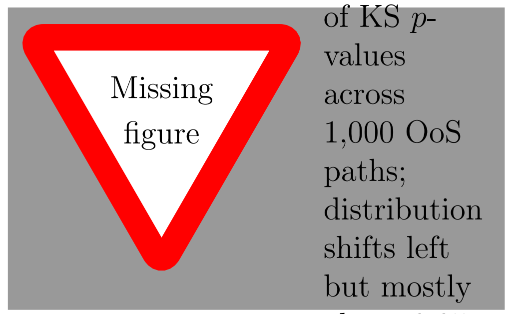


Figure 5: Hyperparameter grid search over (ϵ, λ) for SPY ($N = 100$, 200 paths per grid point). Panel (a): heatmap of the objective function $J(\epsilon, \lambda)$ in $\log_{10}$ scale over the full search grid; the red star marks the optimal pair (ϵ^*, λ^*). The well-defined interior minimum confirms that the jump hyperparameters are identifiable from data. Panel (b): ACF of $|G_t|$ at the optimal parameters computed from 500 jump-containing paths (10th–90th percentile band shaded; mean curve solid navy) compared with the SPY observed ACF (dashed red) and the 95% significance band (dotted black).



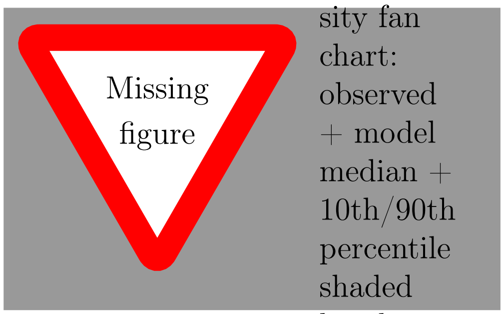
Histogram of KS p-values across 1,000 IS paths; distribution nearly uniform above 0.05

(a) In-sample KS p-value distribution



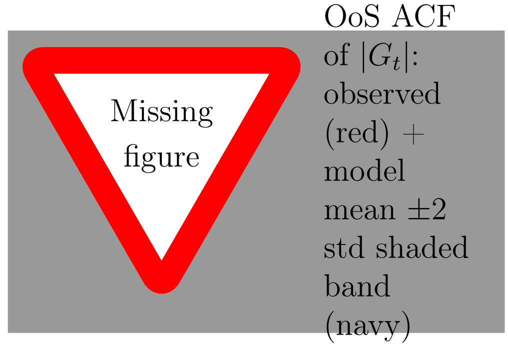
Histogram of KS p-values across 1,000 OoS paths; distribution shifts left but mostly above 0.05

(b) Out-of-sample KS p-value distribution



OoS density fan chart: observed + model median + 10th/90th percentile shaded band

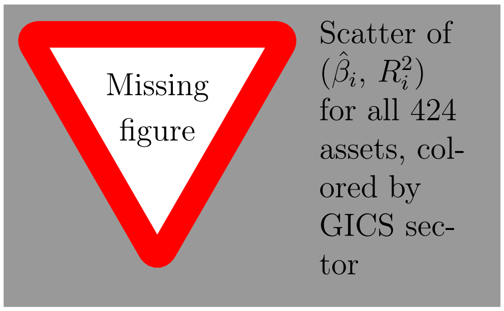
(c) Out-of-sample density fan chart



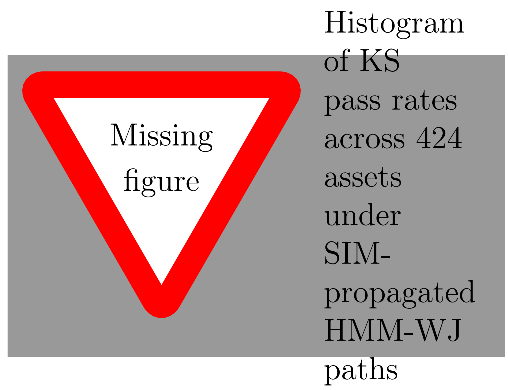
OoS ACF of $|G_t|$: observed (red) + model mean ± 2 std shaded band (navy)

(d) Out-of-sample ACF of $|G_t|$

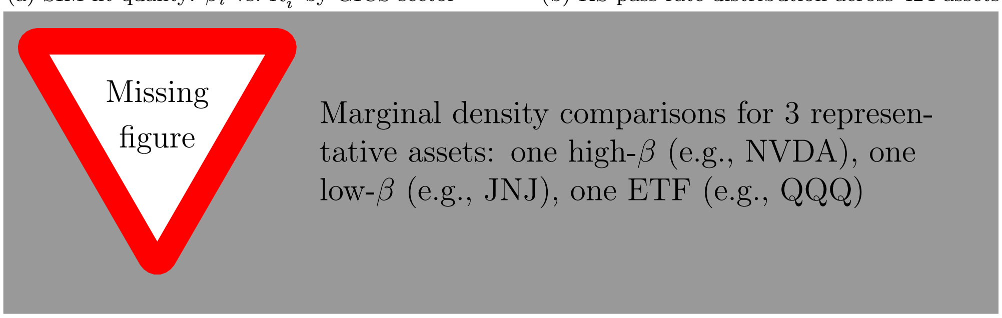
Figure 6: Statistical validation of HMM-WJ ($N = 100$, 1,000 simulated paths, $\alpha = 0.05$). Panels (a) and (b) show the distribution of KS p-values in-sample and out-of-sample, respectively; a well-calibrated generative model produces p-values that are approximately uniform above the significance threshold. Panel (c) shows the out-of-sample marginal density fan chart. Panel (d) shows persistence of volatility clustering in the out-of-sample window.



(a) SIM fit quality: $\hat{\beta}_i$ vs. R_i^2 by GICS sector



(b) KS pass rate distribution across 424 assets



(c) Representative asset density comparisons

Figure 7: Multi-asset extension via the Single-Index Model (SIM) across 424 S&P 500 constituents. Panel (a) summarises the SIM regression fit quality by GICS sector. Panel (b) shows that KS distributional consistency is maintained across the full asset universe when paths are generated via the factor structure $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i G_{\mathrm{SPY},t} + \hat{\eta}_{i,t}$. Panel (c) illustrates representative marginal density comparisons for three assets spanning a wide range of systematic risk exposure.

Table 1: Descriptive statistics and stylized-facts tests for SPY daily excess growth rates (in-sample period 2014–2024; $T = 2{,}766$ trading days). JB = Jarque-Bera normality test. LB = Ljung-Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	**SPY Observed**	**Gaussian**	**Laplace**
Mean (annualized, %)	—	—	—
Std Dev (annualized, %)	—	—	—
Skewness	—	0	0
Excess Kurtosis	—	0	3
JB normality test	reject*	—	—
LB test on G_t (lag 20)	fail to reject	—	—
LB test on $\lvert G_t \rvert$ (lag 20)	reject*	—	—

Table 2: In-sample and out-of-sample model comparison for SPY ($N = 100$, 1,000 simulated paths, significance level $\alpha = 0.05$). GARCH: GARCH(1,1) estimated by quasi-maximum likelihood. HMM-NJ: standard Hidden Markov Model without jumps. HMM-WJ: hybrid model with Poisson jump-duration extension. ACF-MAE: mean absolute error of the ACF of $|G_t|$ over lags 1–252. Kurtosis: mean excess kurtosis across simulated paths. Dashes indicate values pending computation.

Metric	**GARCH(1,1)**	**HMM-NJ** ($N = 100$)	**HMM-WJ** ($N = 100$)
In-sample: 2,766 trading days (2014–2024)			
KS pass rate (%)	6.0	100.0	98.5
AD pass rate (%)	1.4	97.0	94.6
Excess kurtosis (observed)		7.715	
Excess kurtosis (simulated)	8.1	5.7	5.5
ACF-MAE (lags 1–252)	0.030	0.059	0.055
Out-of-sample: 249 trading days (2025)			
KS pass rate (%)	78.1	96.6	94.0
AD pass rate (%)	73.1	96.7	94.2
Excess kurtosis (observed)		6.867	
Excess kurtosis (simulated)	1.6	5.1	4.9
ACF-MAE (lags 1–252)	0.026	0.041	0.040
Parameters estimated	3	2	4

Table 3: Sensitivity of model performance to state resolution N. KS and AD pass rates computed over 1,000 simulated in-sample paths at $\alpha = 0.05$. ACF-MAE = mean absolute error of the ACF of $|G_t|$ over lags 1–252 (pending).

Model	N	**KS pass (%)**	**AD pass (%)**	**ACF-MAE**
HMM-NJ	90	99.4	98.4	—
HMM-NJ	60	99.4	98.3	—
HMM-NJ	30	99.2	98.2	—
HMM-WJ	90	99.5	98.1	—
HMM-WJ	60	99.3	97.6	—
HMM-WJ	30	98.7	97.7	—

Table 4: Multi-asset extension results via the Single-Index Model (SIM) across 424 S&P 500 constituents. Paths generated as $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i G_{\mathrm{SPY},t} + \hat{\eta}_{i,t}$ where $G_{\mathrm{SPY},t}$ is drawn from HMM-WJ ($N = 100$) simulated paths. All statistics pending completion of the 424-asset pipeline.

Metric	**Mean**	**Median**	**5th pct**	**95th pct**
KS pass rate (%)	—	—	—	—
AD pass rate (%)	—	—	—	—
SIM R^2	—	—	—	—
Excess kurtosis match	—	—	—	—

Algorithms

Algorithm 1 Empirical hidden Markov model construction via Laplace partitioning

Require: Price history $P_{1:T}$, Risk-free rate r_f, Number of states N
Ensure: Transition matrix $\mathbf{T}$, Quantile boundaries $\mathcal{Q}$

Compute excess growth rates $R_t = \frac{1}{\Delta t} \ln(P_t/P_{t-1}) - r_f$ for all t.
Fit Laplace parameters (μ_L, b_L) to the series $\mathbf{R}$ using maximum-likelihood estimation.
Define interior boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ for $k \in \{1, \dots, N-1\}$, where F_L^{-1} is the inverse CDF of the Laplace distribution.
Set finite outer bounds $Q_0 = F_L^{-1}(0.001; \mu_L, b_L)$ and $Q_N = F_L^{-1}(0.999; \mu_L, b_L)$.
Assign each excess growth rate R_t to a discrete state $s_t \in \{1, \dots, N\}$ based on the boundaries in $\mathcal{Q} = \{Q_k\}$.
Initialize count matrix $\mathbf{C} \in \mathbb{R}^{N \times N}$ and transition matrix $\mathbf{T} \in \mathbb{R}^{N \times N}$ with zeros.
for $t = 1$ to $T - 1$ **do**
    Identify transition $i = s_t$ to $j = s_{t+1}$.
    $\mathbf{C}_{i,j} \leftarrow \mathbf{C}_{i,j} + 1$.
end for
for $i = 1$ to N **do**
    $\mathbf{T}_{i,:} \leftarrow \mathbf{C}_{i,:} / \sum_{j=1}^{N} \mathbf{C}_{i,j}$ {Row-wise normalization}
end for
return $\mathbf{T}, \mathcal{Q}$

Algorithm 2 Hybrid jump-diffusion simulation with persistent volatility

Require: Model parameters $\mathbf{T}, \bar{\pi}, \epsilon, \lambda, N_{tail}, p_{neg}$, Total Steps M
Ensure: Simulated state sequence $S_{1:M}$

Define tail-states $\mathcal{S}_{bottom} = \{1, \ldots, N_{tail}\}$ and $\mathcal{S}_{top} = \{N - N_{tail} + 1, \ldots, N\}$.
Sample initial state $S_1 \sim \text{Categorical}(\bar{\pi})$.
Set *counter* $\leftarrow 2$.
while *counter* $\leq M$ **do**
  Sample $u \sim \text{Uniform}(0, 1)$.
  if $u < \epsilon$ **then**
    Sample jump-duration $K \sim \text{Poisson}(\lambda)$.
    for $j = 1$ to K while *counter* $\leq M$ **do**
      Sample $w \sim \text{Uniform}(0, 1)$.
      if $w < p_{neg}$ **then**
        $S_{counter} \sim \text{Uniform}(\mathcal{S}_{bottom})$ {Bias toward negative-tail events}
      else
        $S_{counter} \sim \text{Uniform}(\mathcal{S}_{top})$
      end if
      counter $\leftarrow$ *counter* $+ 1$.
    end for
  else
    Sample $S_{counter} \sim \text{Categorical}(\mathbf{T}_{S_{counter-1},:})$.
    counter $\leftarrow$ *counter* $+ 1$.
  end if
end while
return $S_{1:M}$

Algorithm 3 State decoding and continuous excess growth rate reconstruction

Require: Simulated state sequence $S_{1:M}$, Quantile boundaries $\mathcal{Q}$
Ensure: Reconstructed continuous excess growth rates $\hat{R}_{1:M}$

Define $z_{0.999} = 3.09$.
for $k = 1$ to N **do**
    Set $\mu_k = (Q_{k-1} + Q_k)/2$.
    Set $\sigma_k = (Q_k - Q_{k-1})/(2z_{0.999})$.
end for
Initialize the reconstructed excess growth rate vector $\hat{\mathbf{R}}$ of length M.
for $t = 1$ to M **do**
    Retrieve the simulated state $k = S_t$.
    Sample the continuous excess growth rate $\hat{R}_t$ from the normal distribution $\mathcal{N}(\mu_k, \sigma_k)$.
end for
return $\hat{R}_{1:M}$

Supplemental Information

Algorithm 4 Multi-objective grid search for jump hyperparameters

Require: Observed growth rates $\mathbf{R}_{obs}$, ACF max lag $L = 252$, Number of paths $P = 1000$, Time steps $M = 2766$
Require: Grid space for $\epsilon \in [10^{-5}, 10^{-4}]$ and $\lambda \in [1.0, 100.0]$
Require: Kurtosis penalty weight $w_K = 0.20$
Ensure: Optimal jump hyperparameters (ϵ^*, λ^*)

Calculate observed absolute ACF: $A_{obs}(\tau) = \mathrm{ACF}(|\mathbf{R}_{obs}|, \tau)$ for $\tau \in \{1, \dots, L\}$.
Calculate observed global kurtosis: $K_{obs} = \mathrm{Kurtosis}(\mathbf{R}_{obs})$.
Initialize minimum error $E_{min} \leftarrow \infty$.
for each ϵ in grid space **do**
    for each λ in grid space **do**
        Initialize simulated ACF accumulator $\mathbf{A}_{sum} \leftarrow \mathbf{0}$ and kurtosis accumulator $K_{sum} \leftarrow 0$.
        for $p = 1$ to P **do**
            Simulate path using Algorithm 2 to obtain state sequence $S_{1:M}$.
            Decode sequence using Algorithm 3 to obtain continuous rates $\hat{\mathbf{R}}^{(p)}$.
            $\mathbf{A}_{sum}(\tau) \leftarrow \mathbf{A}_{sum}(\tau) + \mathrm{ACF}(|\hat{\mathbf{R}}^{(p)}|, \tau)$ for all τ.
            $K_{sum} \leftarrow K_{sum} + \mathrm{Kurtosis}(\hat{\mathbf{R}}^{(p)})$.
        end for
        Compute ensemble averages: $\overline{A}_{sim}(\tau) = \mathbf{A}_{sum}(\tau)/P$ and $\overline{K}_{sim} = K_{sum}/P$.
        Compute objective: $J = \sum_{\tau=1}^{L} \left(A_{obs}(\tau) - \overline{A}_{sim}(\tau)\right)^2 + w_K \left(K_{obs} - \overline{K}_{sim}\right)^2$.
        if $J < E_{min}$ **then**
            $E_{min} \leftarrow J$
            $(\epsilon^*, \lambda^*) \leftarrow (\epsilon, \lambda)$
        end if
    end for
end for
return ϵ^*, λ^*